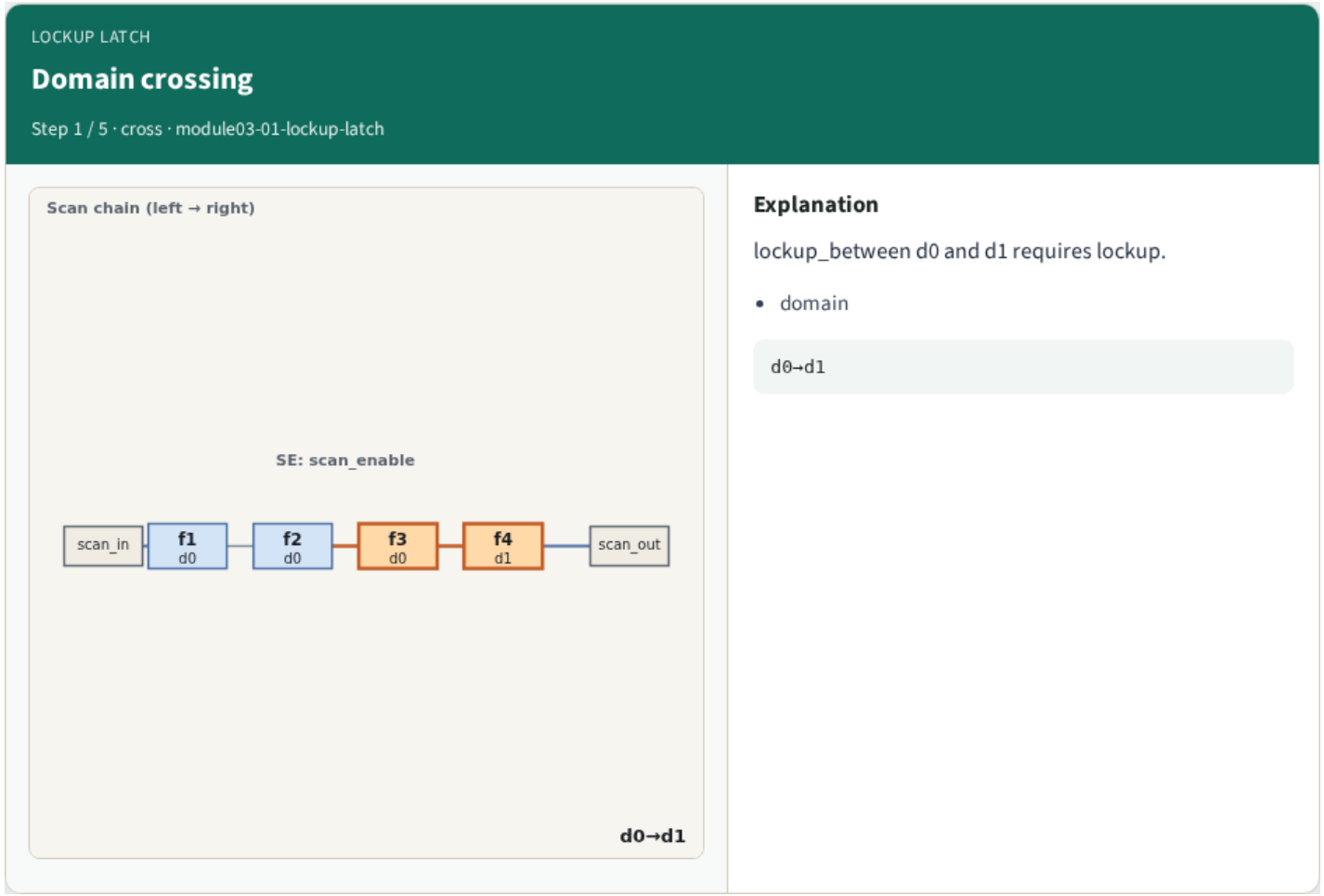
Why lockup

Lockup latches bridge clock domains on scan path

The idea

- GOLDEN: lockup_count=1 on f3→f4 (d0→d1)

Domain crossing



Domain crossing

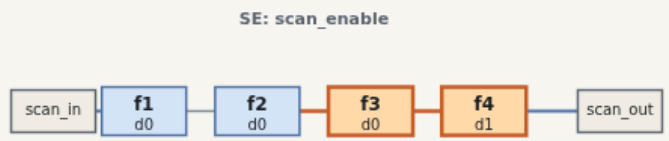
f3→f4 pair

LOCKUP LATCH

f3 → f4 pair

Step 2 / 5 · pair · module03-01-lockup-latch

Scan chain (left → right)



f3→f4

Explanation

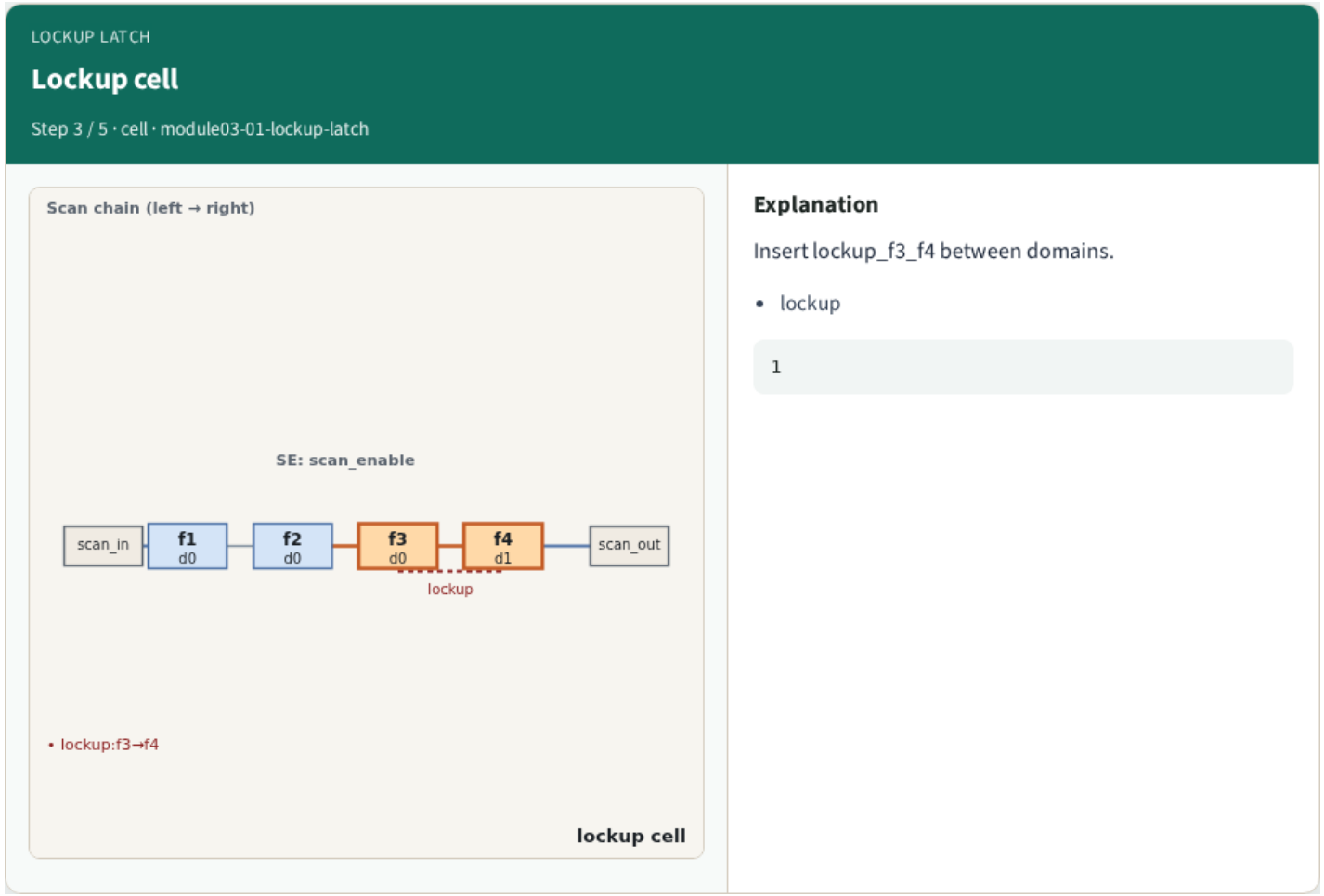
Consecutive flops f3 (d0) and f4 (d1).

- f3
- f4

```
pair
```

f3→f4 pair

Lockup cell



Lockup cell

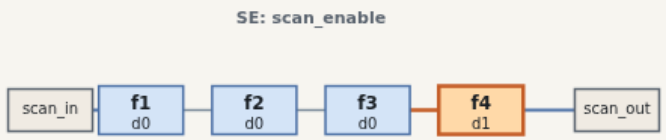
Clock domains

LOCKUP LATCH

Clock domains

Step 4 / 5 · clk · module03-01-lockup-latch

Scan chain (left → right)



Explanation

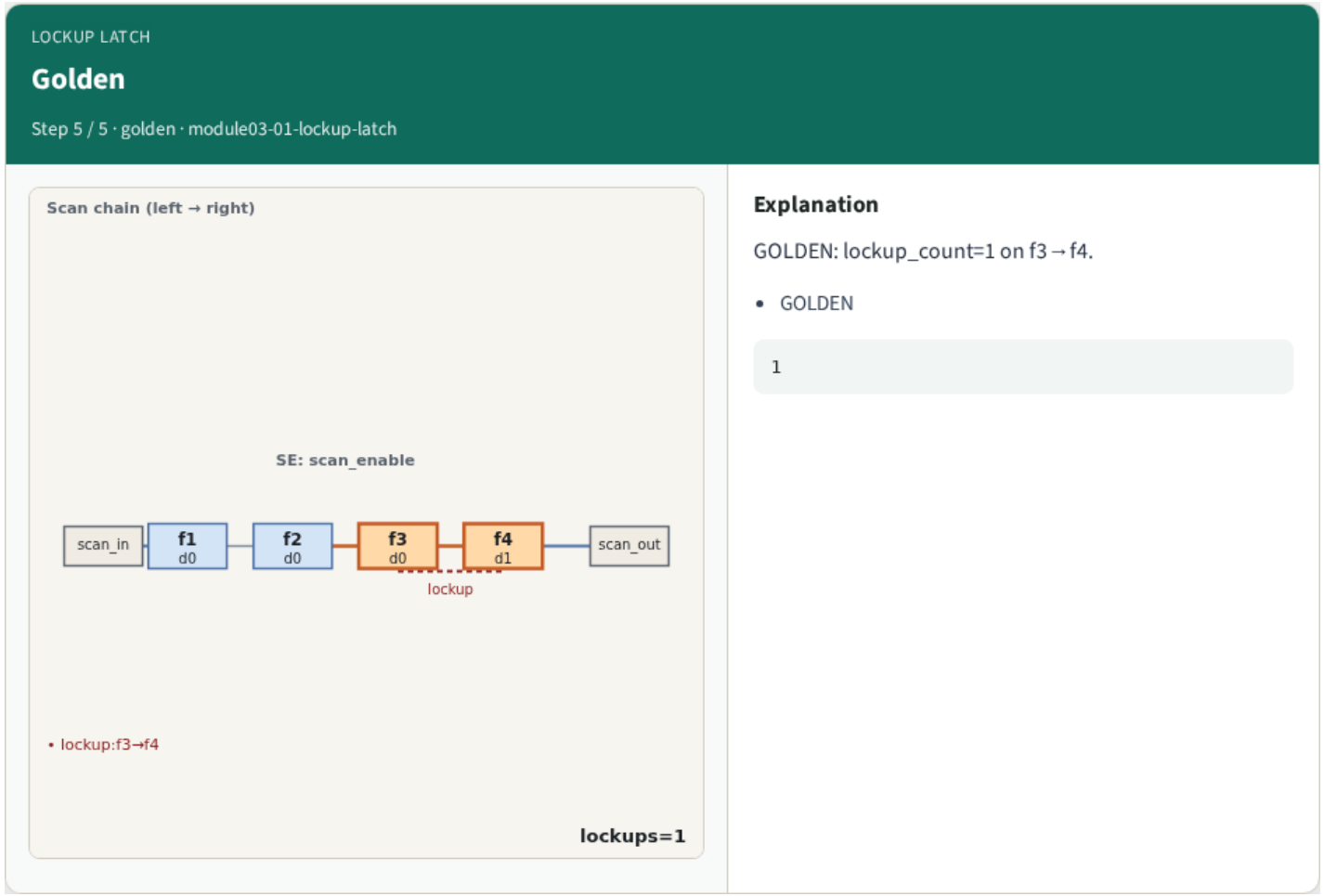
f4 uses clk2; f1–f3 use clk.

- clk

```
2 clk
```

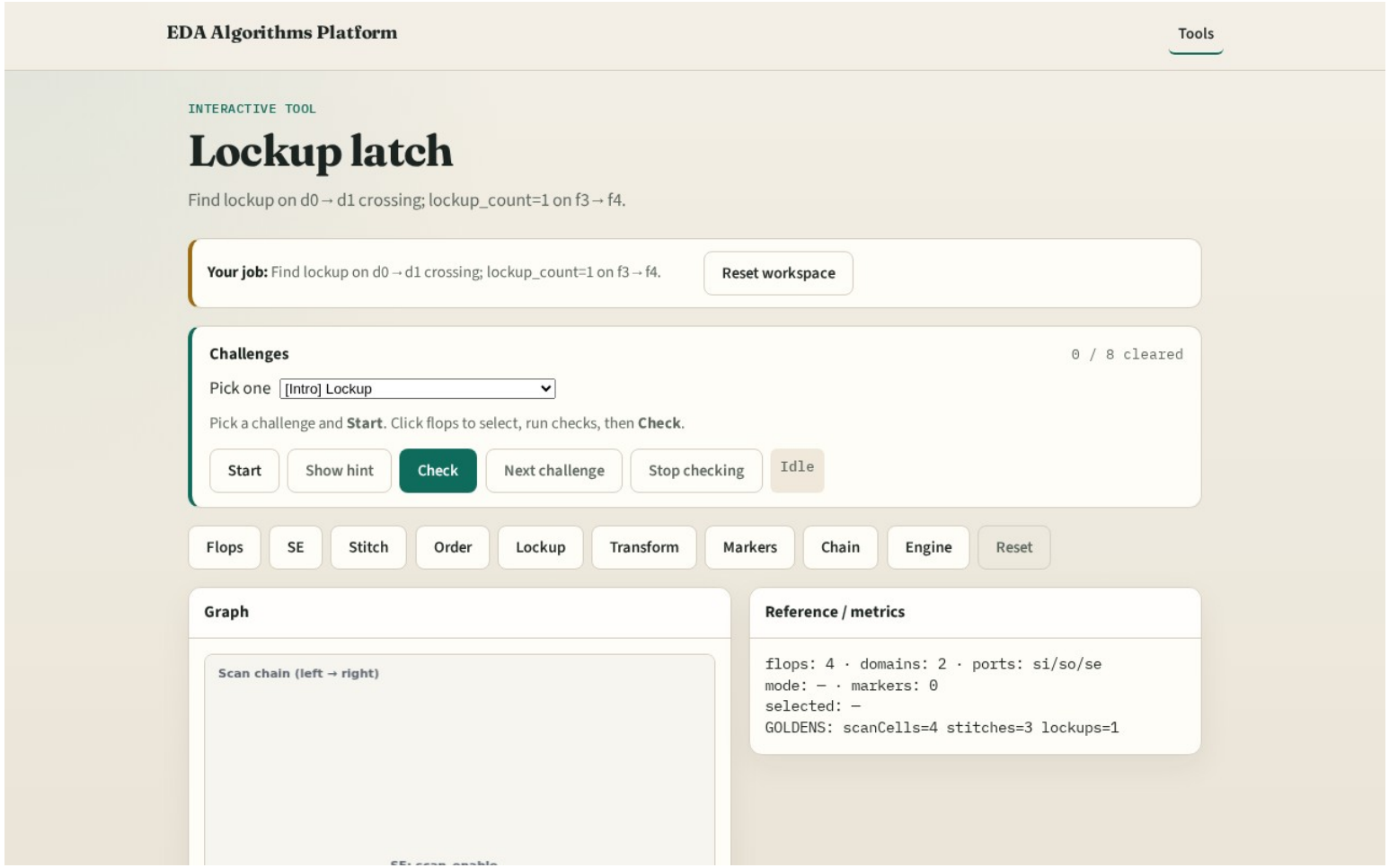
Clock domains

Golden



Golden

Browser lab track



Browser lab starter

Implement track

- Call find_lockups(); assert count==1 and from f3 to f4

Pitfalls

- Lockup on same domain
- Missing clk2 crossing

Your turn

- Next: netlist transform